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MTCAI: Leveraging AI to Simplify Medical Reports for Adults With Intellectual and Developmental Disabilities

Sofia Benson-Goldberg¹ | Ben Satterfield² | Michelle Ramsay²

¹Department of Special Education and Clinical Sciences, University of Oregon, Eugene, Oregon, USA | ²Center for Inclusive Design and Innovation, Georgia Institute of Technology, Atlanta, Georgia, USA

Correspondence: Sofia Benson-Goldberg (sofiabg@uoregon.edu)

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ABSTRACT

Individuals with intellectual and developmental disabilities face significant challenges in accessing and comprehending complex medical documents, contributing to health disparities and unmet healthcare needs. To address this issue an AI-driven tool (MTCAI) was developed and designed to simplify medical texts in accordance with the evidence-based Minimized Text Complexity guidelines. Unlike traditional Plain Language approaches, Minimized Text Complexity guidelines are tailored to readers with limited literacy, including those reading below a third-grade level. This study evaluated MTCAI's effectiveness in simplifying after-visit medical summaries for adults with IDD. The project involved two phases: (1) development and refinement of MTCAI using large language models to systematically automate and apply the Minimized Text Complexity guidelines, and (2) conducting a participatory needs assessment and message testing with five adults with intellectual and developmental disabilities. Participants reviewed medical reports in both their original format and in MTCAI-simplified formats. Quantitative analyses revealed that MTCAI-generated texts significantly reduced reading complexity, achieving levels comparable to those of second-grade texts. Qualitative findings showed that all participants preferred the simplified reports, demonstrated improved comprehension, and expressed a willingness to engage with the simplified documents independently. These results suggest that MTCAI is a promising tool for enhancing health communication accessibility for individuals with intellectual and developmental disabilities. Future research should investigate broader implementation and assess the long-term effects on health outcomes.

1 | Introduction

It is well documented that individuals with intellectual and developmental disabilities (IDD) experience significant challenges in reading comprehension (Wagner et al. 2006; McIntyre et al. 2022). Instead, they tend to be able to read single words and phrases without understanding the content of what they have read (Ratz and Lenhard 2013). Additionally, adults with IDD are known to experience greater instances of co-occurring mental and physical health conditions such as diabetes and obesity than people in the general population (Traci et al. 2002; Tyler et al. 2010). As a result, this group experiences greater unmet healthcare needs and poorer health outcomes (Hanlon

et al. 2018). These health disparities may be partially explained by the complexity of healthcare communication, including physicians' after-visit reports.

The convergence of these two issues (i.e., individual reading comprehension skills and external challenges presented by complex medical documents) presents real complex challenges to the establishment and maintenance of positive health and safety for this population. Individuals with IDD need to establish better understandings of the medical conditions they face. They require more effective means of communication with their personal care physicians. This became clear during the COVID-19 pandemic, as this population was especially vulnerable due to

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their unmet health communication needs (Gleason et al. 2021). The volume and difficulty of the information being produced about COVID-19 caused individuals with IDD stress and confusion (Embregts et al. 2022). Unsurprisingly, adults with IDD reported difficulty accessing information about COVID-19 that they could read and understand independently (Chadwick et al. 2023). This may, at least partially, explain why they were up to eight (Gleason et al. 2021) times more likely to die from contracting COVID-19 than their peers without disabilities.

One way to make texts more accessible to individuals, regardless of their disability status, is to adjust the complexity of the text. Text complexity refers to intrinsic properties of text, including vocabulary, word density, sentence length, and text length. The complexity of a text has correlates with comprehension, with lower complexities resulting in greater reading comprehension scores (Treptow et al. 2007). Given the demands of health documents in terms of vocabulary and syntax, minimizing text complexity is an obvious place to start making these texts more accessible to individuals with IDD.

1.1 | Plain Language Guidelines: An Initial Approach to Text Simplification

Since the passing of the Plain Writing Act of 2010, the United States now requires that all federal agencies use “clear Government communication that the public can understand and use” (1). There are official Plain Language guidelines that accompany the Plain Writing Act (2010) (<https://www.plainlanguage.gov/guidelines>). The legislation and guidelines evolved as a response to national reports indicating a decline in reading ability among people in the United States (see Berkman et al. 2004; Kindig et al. 2004). Going beyond the Plain Writing Act, federal agencies in the United States like the Centers for Disease Control and Prevention (CDC) and the National Institutes of Health have advocated for the use of plain language in health-care generally.

The Plain Language guidelines were developed by a community of federal employees in the United States who were committed to clear communications from the government (PLAIN 2011). The guidelines are broadly organized, offering general guidance on considerations at the word, sentence, and paragraph levels. For example, writers are encouraged to use bulleted lists, tables, illustrations, short words, contractions, and headings to make written text easier to understand.

Research has shown that writing in compliance with the Plain Language guidelines is labor-intensive (Hadden 2015). Writers require explicit training to apply the guidelines. Even with such training, experts may still struggle to apply Plain Language guidelines effectively. This may be because the guidelines are vague and often difficult to interpret (Harper and Zimmerman 2009). As a result, writers must pass materials back and forth with another expert to achieve sufficiently easy texts.

In an attempt to improve the writing process and increase adoption of plain language in health care, several studies have investigated the use of large language models (LLMs) to develop summaries of physicians’ after-visit notes (see Ebby et al. 2025;

Sterling et al. 2024; Tailor et al. 2025). LLMs are computational models trained on extensive text datasets that enable them to interpret, analyze, and generate natural language. The nature and quality of the output are largely dependent on the quality of prompts written by the user. When prompted, LLMs can convert complex information into more accessible formats, potentially making them useful tools for simplifying health care documents. Collectively, this body of research has found that LLMs were able to produce well-crafted summaries that retained pertinent information. However, across studies, limited information has been provided about the prompt(s) used to generate the summaries or the extent to which the summaries matched the Plain Language guidelines.

Although Plain Language guidelines are promising and may result in texts that are readable for many, there are some limitations to applying these guidelines to texts that are intended to be read by individuals with IDD. For example, at best, the guidelines result in texts with readability scores between the sixth and eighth grades. Unfortunately, this results in texts that are still too complex for individuals with IDD (Benson-Goldberg et al. 2024). Furthermore, discrepancies and differences in interpretation and limited agreement among health communication professionals lead to variation in application across writers (Harper and Zimmerman 2009). Given these limitations, there is a clear need to go beyond Plain Language guidelines when simplifying text for individuals with IDD.

1.2 | Minimized Text Complexity Guidelines: A Text Simplification Approach for Individuals With IDD

During the COVID-19 pandemic, in response to the challenge of finding an appropriate set of guidelines to simplify health texts for individuals with IDD, the Minimized Text Complexity (MTC, Erickson et al. 2020) guidelines were developed. Specifically, the development occurred in response to the CDC’s desire to produce health documents that could be read and understood by individuals who read at a third-grade level or below, including those with IDD. Unlike the Plain Language guidelines, which are produced via expert consensus (Harper and Zimmerman 2009), the MTC guidelines were developed based on a rapid review of the literature, resulting in a set of evidence-based guidelines. The guidelines are organized into four sections: (a) whole-text-level guidelines; (b) sentence-level guidelines; (c) word-level guidelines; (d) document-formatting guidelines. Within each section, the MTC guidelines are more specific than Plain Language guidelines. For example, the Plain Language guidelines recommend using short words that most people would understand and avoiding difficult words, whereas the MTC guidelines recommend ensuring that 92% of the words in a passage are among the 3000 most common in the English language. The latter can be calculated using the Text Analyzer tool made available by the Corpus of Contemporary American English (n.d.).

When applied rigorously, the MTC guidelines result in texts that individuals with IDD can read and understand (Benson-Goldberg et al. 2024). Although the MTC guidelines have been folded into the CDC’s Clear Communication Index (Squires,

et al., Squiers et al. 2023), just like Plain Language guidelines, the process remains manually intensive and, as a result, infrequently used. Unlike Plain Language guidelines, there is no published literature investigating using LLMs to produce simplified text using the MTC guidelines. This gap in the literature represents a critical limitation in current text simplification research, as MTC guidelines have demonstrated superior outcomes for individuals with IDD compared to Plain Language approaches, yet lack automation tools for practical implementation.

1.3 | MTCAI

An AI tool called MTCAI was developed to address this gap in implementation. Specifically, the tool uses generative AI to automate the simplification of texts using the MTC guidelines. The goal in designing MTCAI was to create an instrument that produces an accessible and readable summary of a physician's medical report that complies with the MTC guidelines and that an adult with IDD can independently read and understand.

To create MTCAI, a secure internal studio was developed that incorporated LLMs from the Bedrock suite from Amazon Web Services. Guardrails were put in place to prevent sharing of information introduced into the studio environment. This produced a framework which allowed us to examine the interaction of prompts, data (i.e., after visit summaries), and LLM output. The MTC guidelines (Erickson et al. 2020) were used as the basis for all AI prompts within the internal studio described above. Initial testing of the tool was done with healthcare information sheets from prominent open sources (i.e., Medline, NIH, etc.) that addressed health issues common to individuals with IDD were collected and submitted to the MTCAI tool for summarization. LLMs were engaged to produce summaries of these documents using the initial MTC prompt and iterations were made to improve the output. Mock-up medical reports with synthetic data were used to prepare for the summarization of actual after-visit medical reports. Throughout this process, the research team made multiple modifications and adjustments to the original prompt. Each iteration was preserved to track improvements in the output across various LLMs. Outputs from these iterations were monitored to maintain conformity to the MTC guidelines. The outputs of each of the LLMs that were used in the development of the MTCAI tool were tracked and matched to the prompt iterations to determine which combinations of prompts and LLMs consistently produced the most high-quality, MTC-compliant responses.

1.4 | Purpose

The purpose of the work was twofold. First, to establish whether the MTCAI could simplify texts that aligned with the MTC guidelines and expected text metrics. Second, it was to conduct a needs assessment and message testing with individuals with IDD. The purpose of the needs assessment was to understand how individuals with IDD interact with their standard after-visit summaries and to understand what barriers might exist

and to what extent. The purpose of the message testing was to extend the needs assessment into an action phase, where we could begin to understand, with the individuals, whether the individuals in the needs assessment could read and understand their own after-visit summaries when they were simplified with the MTCAI.

2 | Method

The protocol for this project was reviewed and approved by the Georgia Institute of Technology Institutional Review Board. Patients provided permission for the research team to obtain a redacted version of their latest medical report from their physician to be summarized using the MTCAI. The research team partnered with a local medical home organization (a medical practice that incorporates accessible, comprehensive, family-centered support coordination) that focuses on serving adults with IDD. Written consent was obtained from all participants.

2.1 | Participants

Five individuals with IDD were identified from a convenience sample of individuals served by a medical center in the Southeastern US that provides medical services to adults with IDD. The medical home helped with initial contact with potential participants. They issued the invitation to participate to their patients and screened the respondents based on the selection criteria. All participants met the following criteria: (a) at least 18 years of age; (b) currently residing in the United States; (c) have a diagnosis of intellectual or developmental disability; (d) ability to read connected text; and (e) ability to communicate their preferences (including with augmentative and alternative communication). Of the five participants, two had primary diagnoses of Autism and three had primary diagnoses of Down Syndrome. Two were female and three were male. They ranged in age from 24 to 30 ($M = 25.6$). The participants had mixed interest in reading. One (20%) participant indicated that they read books, one (20%) indicated they only read if they are interested, two (40%) reported they enjoy listening to audiobooks, and one (20%) said that they have not been reading lately.

2.1.1 | Application of the MTCAI to After-Visit Summary Reports

The team at the medical home redacted the after-visit summaries. The redacted summaries were then reviewed for information that was relevant to the participants. It was decided to focus on the narrative summaries, which had the headers of *History* and *Instructions* in the original reports. Once they were extracted from the summary, these sections were then processed through the MTCAI to develop simplified documents to be used during message testing. The first author led the simplification process. All simplified reports were reviewed by a physician at the medical home to ensure accuracy. This resulted in five personal, simplified texts. The simplified reports were then compared to the original reports on several key metrics related to readability: (a) total words; (b) words per sentence; (c) Flesch Reading Ease scores; (d) Flesch-Kincaid

TABLE 1 | Example of report simplified with the MTCAI.

Initial report	MTCAI summary
<i>Medical Update</i>	<i>Health Update</i>
Client presents for his annual health maintenance exam and evaluation at The Practice clinic. He is living at home with his parents. He is an active member of the household. He is volunteering at the Local public library (one session per week) and works at the coffee shop at his church twice a week.	You have Down Syndrome. You live with your parents. You help at home. You work at a library. You work at a coffee shop.
He has had difficulty sustaining his healthy lifestyle and weight loss program. He has gained 12 pounds since his last visit. He is exercising but not as much as recommended. Recommend continuing with the Local Fitness program. He has a trainer once a week at Local Gym and is considering increasing to twice weekly. We discussed weight loss options—including use of GLP-1 s; He meets BMI criteria for Wegovy therapy and has no contra-indications. Plan is to continue strict, calorie reducing diet for 3 months, and then consider adding medications if no improvement. He is due for the following health maintenance studies: urinalysis, lipid profile, BP and BMI.	<i>Weight Gain</i> You gained 12 pounds. You need to exercise more. You go to the gym once a week. You may go twice a week. You need to eat less food. You may take a medicine to help you lose weight.
His immunizations are not up to date. We recommend COVID booster. Mom prefers to wait for new booster in the Fall. Recommend Prevnar 20—will update today. Recommend seasonal flu shot in the Fall.	<i>Immunizations</i> You need some shots. You need a COVID shot. You need a Prevnar 20 shot. You need a flu shot in the fall.
Patient has a history of severe obstructive sleep apnea; He has not been able to tolerate CPAP. His sleep apnea symptoms were much improved when he managed to get his weight down into the 180 range. He manages well with sleeping on his side. We recommend restarting slumber bump pillow. Occasional daytime napping. He needs follow up with ENT regarding sleep apnea management options.	<i>Sleep Apnea</i> You have sleep apnea. You had trouble using a CPAP machine. You sleep better on your side. You use a special pillow. You need to see an ENT doctor.

grade-level score; and (e) the percentage of words that appear within the 3000 most frequently occurring words. See Table 1 for an example of an initial and simplified report; to protect the anonymity of the participants, the original report is a synthesis of several initial reports.

2.2 | Procedures

Consenting participants scheduled two virtual appointments with the second and third authors. Both authors were present for both sessions with all participants. During the first session, the participants engaged in a participatory needs assessment to understand their previous experiences and current needs related to reading and understanding their after-visit summaries. The second session was intended to message test the simplified summaries. Descriptions of both sessions are outlined in more detail below. All sessions were recorded and automatically transcribed via Zoom. Participant statements were manually transferred to Microsoft Excel for analysis (Amozurrutia and

Servós 2010). All coding and analysis were done by the second and third authors.

2.2.1 | Participatory Needs Assessment

While MTCAI was being developed and applied to the after-visit summaries, we completed a participatory needs assessment with the participants. We used a semi-structured protocol to guide the participatory needs assessment to allow us to gather information about participants' previous experiences with their medical reports as well as their needs around understanding their medical reports. Participants were asked to review their original after-visit summary. We asked participants to identify sections of their reports (i.e., diagnosis, physician instructions, medications, etc.) and to discuss that aspect of their lives. We asked each to read the narrative section that described their history (i.e., a section that was later simplified) then asked questions to ascertain comprehension. As a part of this we asked the participants to identify any difficult words

they encountered. See Table S1 for the participatory needs assessment interview guide.

2.2.2 | Message Testing

After the participatory needs assessment was completed, participants scheduled a follow-up appointment to complete the message testing. During this session, the simplified version of their medical report was presented to them. Again, the participants were asked to review the report and answer a variety of questions aimed to understand their comprehension of the report as well as impressions of the report. The questions used were modeled after those used in the first interview. Again, we asked the participant to read a narrative section and to identify difficult words. Lastly, we asked the patient to tell us which version of their report they preferred and why. See Table S2 for the message testing interview guide.

3 | Results

The results of the study provide preliminary evidence that the MTCAI can be used to simplify medical reports and produce summaries that comply with the MTC guidelines and that key metrics suggest the simplified reports are easier to read than the original reports. Furthermore, results of the needs assessment suggested that this type of simplification is necessary and results from the message testing indicated that the simplified resulting reports were preferred by the participants.

3.1 | Comparing the Original and Simplified Reports

When analyzed as a group, the original physician's reports for these participants were written at an 8th grade reading level on average ($SD = 1.11$), with a low Flesch Reading Ease score (53.2) indicating they were quite difficult to read (Flesch Reading Ease scores range from 0 to 100 with higher scores

indicating texts are easier to read). The original reports used long sentences with complex syntax and unfamiliar words that are infrequently used in written English. In contrast, the metrics for the MTCAI summaries of these reports indicated they complied with the recommended MTC guidelines, providing shorter sentences and less complex words that are more frequently occurring in written English. The simplified reports were written at a 2nd grade reading level on average ($SD = 0.44$) and had a higher Flesch Reading Ease score (88.4), suggesting they were easy to read.

A series of *t*-tests were run to evaluate differences in (a) total words; (b) words per sentence; (c) Flesch Reading Ease scores, (d) Flesch–Kincaid grade-level score; and (e) the percentage of words that appear within the 3000 most frequently occurring words. The first four metrics were taken directly from Microsoft Word, while the last metric was measured using the Corpus of Contemporary American English (n.d.). After visual inspection of boxplots, no extreme outliers were found. Shapiro Wilk's test revealed the data were normally distributed, so analyses were conducted on all variables. A one-tailed paired *t*-test revealed significant differences (i.e., $p < 0.05$) for all variables with large effects sizes (i.e., Hedges $g > 0.8$). This suggests that the MTCAI were indicative of texts that are significantly easier to read than the original documents. See Table 2 for the average metrics across each document type (i.e., original and simplified), the tools used to obtain each metric, the mean difference between the metrics across texts, and the *p* values and Hedges *g* values from the paired sample *t*-tests.

3.2 | Participatory Needs Assessment Results

Participants were asked to review their original physician's reports and locate specific information on their reports. Only two participants (40%) reported having previously read one of their reports. These patients received their report directly from their doctor by mail following the visit. These two expressed interest in reading their reports. The other participants (60%) said they did not personally receive a copy of their previous reports

TABLE 2 | Average metrics for original and simplified documents.

Metric	Original mean (SD)	Simplified mean (SD)	Mean difference	Paired sample <i>t</i> -test	
				<i>p</i>	Hedges's <i>g</i>
Total Words ^a	522.2 (208.19)	347.4 (87.28)	174.8	0.041 ^c	212.19 ^d
Average number of words per sentence ^a	10.3 (1.98)	5.72 (0.7)	4.58	0.006 ^c	2.90 ^d
Flesch Kincaid reading ease score ^a	53.2 (5.3)	88.4 (2.0)	35.2	<0.001 ^c	7.20 ^d
Grade level equivalent on readability index ^a	8.12 (1.11)	2.3 (0.44)	5.82	<0.001 ^c	1.67 ^d
Percent of words among 3000 most frequently occurring in English ^b	79.6% (2.3%)	90% (1.7%)	10.4%	<0.001 ^c	3.39 ^d

^aMetric generated using Microsoft Word.

^bMetric generated at <https://www.english-corpora.org/coca/>.

^cIndicates a statistically significant difference.

^dIndicates a large effect size.

and said they were not sure they wanted to read their reports. All participants said they looked to their parents for medical information.

Upon reviewing their report, two of the participants (40%) said that they thought it was important for them to read their report. Four (80%) said they understood some of what they read in their report, but all five described the report as “hard” and “confusing” and one participant noted, “there were some big terms in there.” Another participant made a comment about the physical layout of the document, saying, the “font is too small [...] too many words too close together.” All five participants identified at least five words that they did not recognize and identified unfamiliar acronyms.

When asked to review their report, three participants (60%) found the section with their doctor’s instructions without assistance. Only one (20%) was able to locate information about their diagnosis without assistance. All five reported being unsure that they understood the doctor’s instructions. When asked what the report said about their diagnosis, three (60%) could make accurate statements. However, none of their comments were reflective of the specific statements and information contained in the report, but rather general information about the diagnosis.

When asked what could make the report better participants suggested reducing the amount of information (i.e., fewer words) and better organizing the document, including using formatting features (e.g., bold headers), grouping similar information together, and ensuring all information is current. For example, one participant said,

Yes, there’s a lot of words on each page. [...] It would help if there was less information. It would help if the words were a little more easy to understand, or maybe smaller or shorter.

Three participants commented on the format, one indicating that the font needed to be bigger, and the other two commenting on organization. For example, one participant pointed to several important topics in their report that were all in the same paragraph and recommend they be:

boldened or have those boxed. Because those are good for people to know. And so, it’s not all running all together. It can just be separate for them. Or have it bolded or something to make it just more stand out.

Similarly, one participant suggested grouping like information, saying, “it should be more separate [...] they’re not necessarily grouped together in a way that makes a lot of sense.” An unexpected finding was when one patient suggested that the reports should be reviewed for relevancy and recency. They commented that:

a lot of what is in the report is old. Like I don’t take some of the stuff anymore, but it is still in the report.

Some of my other doctors had things that were left on from, I don’t know, since I was a child.

3.3 | Message Testing Results

Following the needs assessment, a second interview provided the participants with a chance to review the MTCAI simplified versions of their physician’s reports. When first shown the new summaries, the participants were asked what they noticed about the summary. All five participants reported that they could read the summary without help. One participant noted, “I can read this one better. Yes, can read the new one better.” All five indicated that they would try to read the simplified report if their doctor gave it to them. All five participants indicated a reduction in the number of unfamiliar or difficult words. One participant noted, “the language seems different [...] I kind of like that [*reads a line from the report*] like that is pretty basic.”

Regarding comprehension, all five patients made correct statements regarding the doctor’s specific instructions to them (e.g., “The doctor says he wants me to put on sunscreen”). All five were able to make statements that suggested correct inferences regarding the doctor’s general instructions to them (e.g., when they read that they need to keep exercising, they inferred “I need to keep riding my bike”). One patient went beyond this and after reading made two independent decisions about recommendations from their doctor to help alleviate foot pain. The report indicated they could use (a) either a regular or medicated pad and (b) different shoes. They shared, “I want use a medicated pad, yes, I may change shoes.” Similarly, another participant was able to express a preference for having necessary shots on different days, as opposed to all on the same day.

Three individuals (60%) volunteered comments that the MTCAI version of their report was organized in a way that was easy for them to follow. Two (40%) mentioned the length of the document compared to the original, stating they preferred it because there were “a lot less papers to go through” and “I think this is much better of having to read this than it is for the 20 page report that they give you each time.” Another reported, “this is so much easier for me in particular. Everything on here is short and sweet and not too, not like a big analyze [...] which I really appreciate.” That same participant suggested that they have a couple of friends who reading “is not easy for them” and that “this might be a tool that might help them.”

When asked their preference, all five indicated that they would rather read the MTC summary than the original physician’s report and that they would read the simplified version if it were given to them by their doctor.

4 | Discussion

Just as LLMs have been found helpful to produce Plain Language summaries (see Ebby et al. 2025; Sterling et al. 2024; Tailor

et al. 2025), we found that LLMs can be used to create simplified reports that comply with the MTC guidelines. Additionally, the metrics suggest that the resulting simplified reports are significantly easier to read than the original medical reports. The metrics of the original reports suggested the documents were too challenging for adults with IDD. Additionally, the results of the needs assessment demonstrated a need for simplified reports. Results from the message testing demonstrated that the participants preferred the summaries produced by the MTCAI, believing they were easier to read and better organized. Their interactions while reading the simplified summaries suggest better comprehension that in some cases facilitated decision making.

The needs assessment demonstrated a critical need for simplified after-visit summaries. As is consistent with the literature (Roch and Levorato 2009), the adults in the study were able to read and identify most words in the original reports. However, they had difficulty comprehending what they read. Even when they were able to make accurate statements about their diagnoses, their statements did not align with those made in the reports by their doctors. This suggests that they were relying on prior knowledge to answer the questions rather than comprehending the specifics in their reports.

Although unsurprising, it seems problematic that most participants reported never having received a copy of their after-visit summary before. Without access to information about their conditions, it is difficult to imagine how individuals with IDD might partake in supported decision making about their own health. Future research is warranted to investigate why individuals are not receiving their reports and to determine the extent to which access to simplification tools, such as the MTCAI, may increase the likelihood that individuals with IDD are given access to this critical information. Future research should investigate how this access might be used to engage in supported decision making.

The results of the message testing are encouraging, suggesting that the simplified reports could be read and understood by the participants. The unanimity of the responses was unexpected, with all participants indicating that they could read the report without help, preferred the simplified texts, and would read the report if it were given to them by their provider. This is important because perceptions of texts impact whether they are read and how they are understood. Future research is warranted to expand the scope of the current study. If these results hold true, they suggest that the MTCAI creates texts that are both attractive and useful for individuals with IDD. Perhaps most importantly, all participants made comments that were correct statements directly related to their reports.

5 | Limitations and Future Considerations

The foremost limitation of this preliminary study is the small sample size, which consisted only of individuals with Down syndrome and Autism. This makes it difficult to extend the findings to the larger population of individuals with IDD. Future work is needed to understand the extent of the need for simplified after-visit summaries as well as the impact of summaries simplified

with the MTCAI on reading comprehension for a more representative sample of adults with IDD. Further research should go beyond reading comprehension to investigate whether increased comprehension leads to increased adoption of medical recommendations. Similarly, future research should investigate whether access to simplified reports supports increased participation in supported decision-making.

Another limitation is that we did not measure individuals' reading comprehension scores. This also limits our ability to make inferences about how the findings relate to the population of individuals with IDD. Though there was little evidence that the group as a whole enjoys reading or spends much time reading, the results could indeed be impacted by their baseline reading comprehension skills. Future work should consider measuring participants' reading comprehension using standardized assessments to understand the impact of reading comprehension on outcomes.

Lastly, the study is limited by the exploratory nature of the design. Future work should consider employing more rigorous methodologies to understand the impact of simplified texts on individuals' ability to read and comprehend their after-visit summaries. Additionally, our approach does introduce potential bias, as the researchers who conducted the needs assessment and message testing were not blinded to the purpose of the study and were involved in the development of MTCAI. Future work should consider employing research assistants who are blinded to the purpose of the study to code and analyze the participant data.

5.1 | A Note on Potential Dangers of GenAI in Medical Settings

MTCAI was developed for the purpose of simplifying medical health documents. With that in mind, precautions were taken to ensure the security of participants' private health information. First, a secure internal studio was developed with guardrails in place to prevent the sharing of information beyond the studio environment. This means that none of the data entered into MTCAI were shared back out to the LLMs that were used. Second, all the original reports viewed by the researchers were redacted by the medical health home. This further ensured that no identifying information was shared with MTCAI. Future research and development endeavors need to continue to consider the implications and ethics of using LLMs and generative AI in the context of the medical field. Of particular concern is the privacy of private health information. Additionally, there are concerns around the accuracy of the output. LLMs are known to hallucinate. Given the importance of these documents, it is important for all efforts to include human-in-the-loop review, meaning no AI output goes directly to patients or anybody else without first being checked by a human. In the case of research, this may be a member of the research team or a content area expert. In clinical practice, this would mean the care provider who is creating the simplified text.

6 | Conclusions

This preliminary study demonstrates the need for simplified medical health documents for adults with IDD and the

potential of the MTCAI tool to provide this increased accessibility. By automating the application of the MTC guidelines through LLMs, MTCAI successfully transformed complex after-visit summaries into simplified texts that participants could read and understand. Additionally, the findings from both quantitative readability metrics and qualitative message testing underscore the tool's effectiveness in reducing barriers to health communication. All participants preferred the simplified reports and showed improved comprehension. Although the study is limited by its small sample size and exploratory design, the results are encouraging and warrant further research. Future work should explore broader implementation, assess long-term impacts on health outcomes and health literacy, and investigate how tools like MTCAI can support informed decision-making and equitable access to health-care information.

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Ethics Statement

All study procedures were approved by the Institutional Review Board at Georgia Institute of Technology.

Conflicts of Interest

Sofia Benson-Goldberg was a consultant on the development and application of the MTCAI tool and led manuscript preparation efforts including initial drafting and data analysis. Ben Satterfield was the principal investigator on the project and a contributing writer, including reviewing drafts and data analysis. Michelle Ramsay was a project team member and manuscript reviewer. The other authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Participatory needs assessment protocol. **Table S2:** Message testing interview protocol.